ALICE L WALTON SCHOOL OF MEDICINE

Dyslipidemia: Pharmacological and Non–Pharmacological options

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Authors: Diptiman Bose, Phil Riley, Trager Hintze
Editors: Jordan Finn, Russell Johnson

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Learning Objectives (3)

After completing this brick, you will be able to:

1. Justify the use of antihyperlipidemic agents based on their mechanism of action and pharmacokinetic profile
2. Recognize common adverse drug reactions for antihyperlipidemic agents
3. Discuss the non-pharmacological options to treat dyslipidemia

CASE CONNECTION

Mr. L, a 58-year-old man with a history of hypertension, hyperlipidemia, and type 2 diabetes, presented to the outpatient clinic with a three-week history of diffuse muscle aches, mainly in his thighs. He described the pain as persistent, dull, and achy, without associated weakness, trauma, or dark urine. He was recently started on atorvastatin 40 mg daily two months ago, in addition to lisinopril and metformin. He denied alcohol, tobacco, or illicit drug use. Mr. L did not report any fever, rash, chest pain, or dyspnea.

Mr. L is concerned about why he has this pain in the muscles. He said he is also looking at other ways to lower his cholesterol. How would you address Mr. L's concerns? Are there other ways to lower cholesterol?

GO TO CONCLUSION ↓

What are antihyperlipidemic drugs? (LO#1 and 2)

Antihyperlipidemic agents lower blood levels of lipids, including low-density lipoprotein (LDL) and cholesterol. They are used to treat lipid disorders (dyslipidemia), which often present with high levels of cholesterol and/or triglycerides (TGs). These disorders are most often seen in otherwise well patients but can also be secondary to conditions like diabetes mellitus or renal or liver disease or more rarely from genetic abnormalities of lipid metabolism.

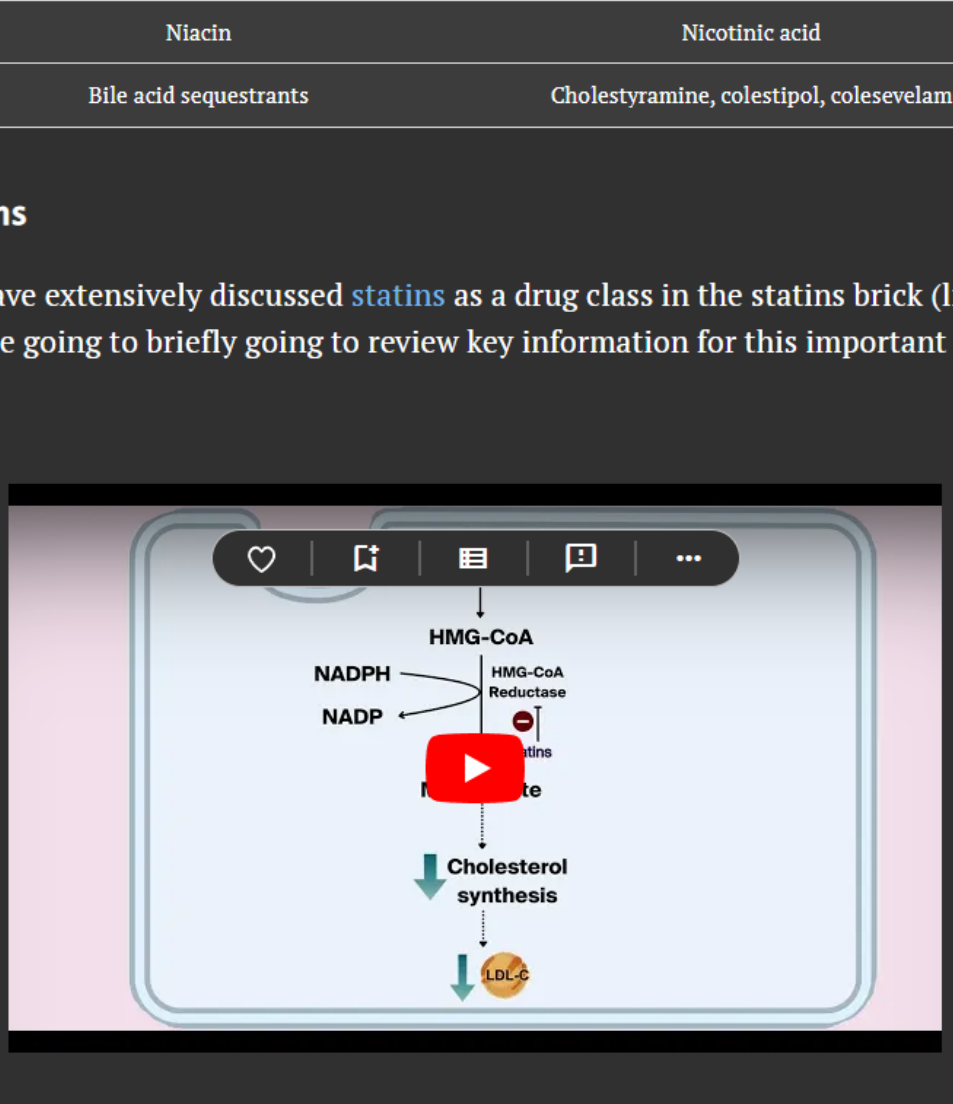
The table below lists the agents used to treat dyslipidemia. We have discussed statins in detail in th an classify these agents based on their be discussed on their benefits to reduce atherosclerotic cardiovascular disease (ASCVD) risk. This includes the risk of reducing myocardial infarction, stroke/ischemic attacks, angina, and peripheral arterial disease (PAD). Let's discuss each drug class in further detail

Table Anti-lipid agents

Mechanism of action/Drug class	Examples
HMG-CoA reductase inhibitors/statins (atorvastatin, simvastatin)	Atorvastatin, Simvastatin
Cholesterol-absorption inhibitors	Ezetimibe
PCSK9 inhibitors	Alirocumab, Evolocumab
PPAR-agonist/Fibrates	Gemfibrozil, Fenofibrate
Fish oil and marine omega-3 fatty acids	EPA, DHA, α-linoleic acid
Niacin	Nicotinic acid
Bile acid sequestrants	Cholestyramine, colestipol, colesevelam

Statins

We have extensively discussed statins as a drug class in the statins brick (link). We are going to briefly going to review key information for this important drug class.



Lipid-lowering has been proven to improve survival in patients with hypercholesterolemia for primary prevention (no clinical evidence of coronary disease) and secondary prevention (with established coronary disease). This benefit has been shown when cholesterol levels are “normal” or even borderline “high.” While lowering lipid levels can explain these benefits, other factors also play a role.

- **Plaque stabilization**: Coronary plaque rupture can lead to coronary syndromes. Lipid lowering can contribute to maintaining the integrity of the fibrous cap of the plaque, thus protecting rupture.
- **CRP reduction:** Statins can decrease CRP levels, thus interfering with inflammatory response. The mechanism of this reduction is not clear.
- **Reversal of endothelial dysfunction:** The vasoconstriction associated with endothelial dysfunction is reduced with statin therapy, however the mechanism underlining this effect is not clear.
- **Regression of a** sclerotic lesions occurs with statins without any changes to the vessel thickness. Exact timing is not clear, but some studies have reported occurrence within 12-24 months of therapy

How do statins contribute to plaque stabilization in coronary arteries?

What are potential ADRs from Statin use?

- ***Hepatic dysfunction***: The majority of statins are taken orally and have low systemic bioavailability since they undergo extensive first-pass metabolism by the liver. Thus, monitoring liver function is important with statin use. Statin can elevate liver aminotransferases in patients receiving statins within the first three months of therapy and is dose-dependent. A LFT is usually ordered before initiating statin therapy and is repeated only after clinical indications. Lower statin dose or using alternate agents is recommended in levels with a confirmed second
- ***Stain-associated muscle symptoms (SAMS):*** SAMS is one of the most common adverse effects of statin use. SAMS can present as myalgia (muscle aches, soreness, tenderness, or cramps), myopathy (muscle weakness ± CPK elevations), myositis (inflammation), or clinical rhabdomyolysis (significantly elevated CPK levels with myoglobinuria and acute kidney injury). Risk of myopathy is lowest with Fluvastatin, pravastatin, and pitavastatin since they are not CYP3A4 metabolized. On the other hand, SAMS has a higher risk with the use of Lovastatin, Simvastatin, and Atorvastatin. Simvastatin dose is maxed at 40mg/day to prevent SAMS. Hypothyroidism can also predispose patients for SAMS. Low vitamin D levels are associated with high rates of myalgia and myopathy.
- ***GI effects***: Diarrhea is a common adverse effect of the drug class.
- ***Diabetes mellitus***: Statin therapy can cause an increased risk of developing diabetes by affecting glucose metabolism in nondiabetics or interfering with glycemic control in patients with diabetes. Just as statins can increase LDL-R in the liver and increase uptake of LDL-C in the liver, the same occurs in the pancreas. This can increase the chances of lipotoxicity in the β-cells resulting in decreased insulin secretion and loss of glycemic control.
- ***Risk in Pregnancy and breastfeeding***: Statins should be discontinued in pregnant patients and breastfeeding is not recommended in patients taking a statin.

What are the potential drug-drug interactions with statins?

Simvastatin, lovastatin, and atorvastatin are metabolized by CYP3A4. Fluvastatin clearance is partly dependent upon CYP2C9 metabolism. Rosuvastatin, pitavastatin, and pravastatin are cleared primarily by non-CYP450 transformati can increase the risk of statin myopathy when administered with statins extensively metabolized by CYP3A4. Drugs such as glucocorticoids, daptomycin also increase the risk of myopathy.

Calcium channel blockers –Calcium channel blockers amlodipine, diltiazem and verapamil are moderate inhibitors of CYP3A4 metabolism and can increase side effects to statins metabolized by CYP3A4.

Grapefruit juice—Grapefruit juice inhibits intestinal CYP3A4, and daily consumption of the juice with Simvastatin increases its level. Rosuvastatin, pravastatin, and pitavastatin is recommended for patients who consume large amounts of grapefruit or grapefruit juice.

Ezetimibe

How does Ezetimibe work?

Ezetimibe's mechanism of action involves the inhibition of cholesterol absorption. The inhibition occurs in the brush border of the intestine. Ezetimibe inhibits the Niemann-Pick C1 like 1 (NPC1L1) protein. NPC1L1 is expressed both in the intestine and the liver, and inhibition of this protein results in a net decrease in cholesterol absorption from the intestine.

Ezetimibe is used alone or in combination with simvastatin. The effect of lowering LDL levels varies between 17-36% and TGs by 5-10%. It can increase HDL levels by 1-3%.

What are the potential ADRs?

Myalgias, diarrhea, URTIs (Upper Respiratory Tract Infections) are some common adverse effe

Bempedoic acid

How does Bempedoic acid work?

The adenosine triphosphate citrate lyase (ACL) enzyme converts citrate to Acetyl CoA, a precursor for HMG-CoA. Bempedoic acid inhibits adenosine triphosphate citrate lyase (ACL). The acetate and Acetyl CoA are important precursors for cholesterol formation. Thus, bempedoic acid can decrease cholesterol formation and LDL levels, leading to upregulation of LDL-C receptor levels. The upregulation of LDL-C receptors subsequently decreases circulating LDL-C levels and uptake of LDL-C by the liver.

What are the potential ADRs?

Hyperuricemia (26%) (used with caution in patients with gout).

What enzyme does bempedoic acid inhibit?

Niacin

How does Niacin work?

Lipolysis in the adipose tissue helps convert TGs to fatty acids. The liver uses the circulating fatty acids for TG synthesis and production of VLDL. Triglycericdes are converted to fatty acids by lipolysis. In the adipose tissue, niacin decreases cAMP function, which decreases the formation of fatty acids. The decrease in circulating fatty acids decreases hepatic VLDL production and, subsequently, LDL-C levels. The decrease in TG is 20-50%, and LDL-C levels decrease by 5-25%.

Niacin has poor tolerability and is rarely used in patients with hypercholesterolemia.

What are the potential ADRs?

Flushing: Intense cutaneous flush

Hepatotoxicity: Niacin can cause severe hepatoxicity and is contraindicated in patients with hepatic disease

CLINICAL CORRELATION

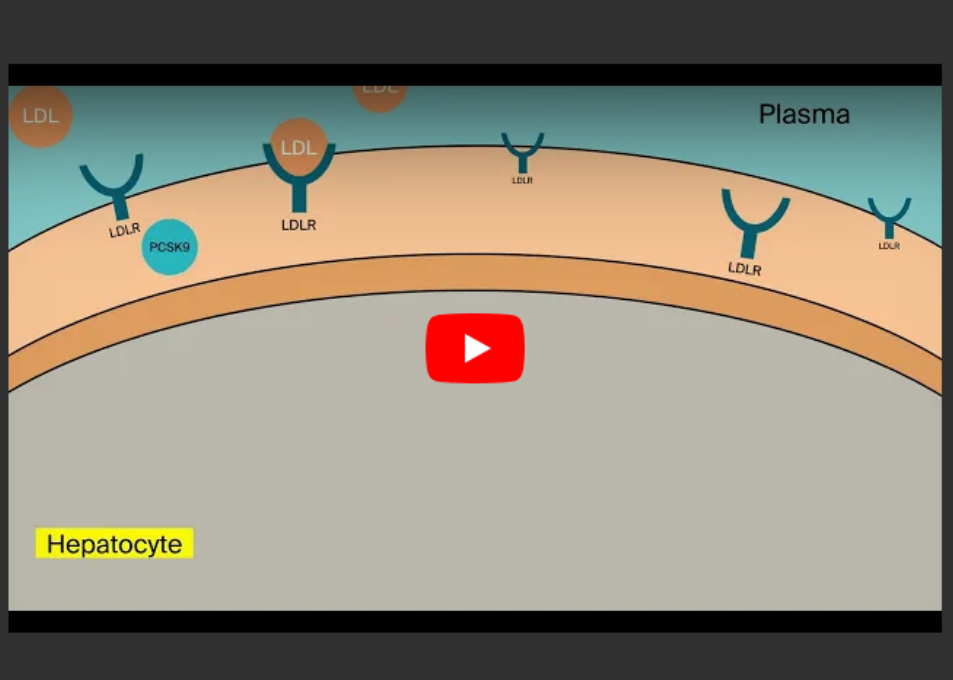
Niacin can cause intense cutaneous flush accompanied by an uncomfortable feeling of warmth and pruritus. Use of extended-release formulation rather than immediate release can decrease flushing. The flushing is prostaglandin mediated; therefore patients can take an aspirin (30 minutes before) (PG inhibitor) prior to the use to stop flushing.

PCSK9 inhibitors

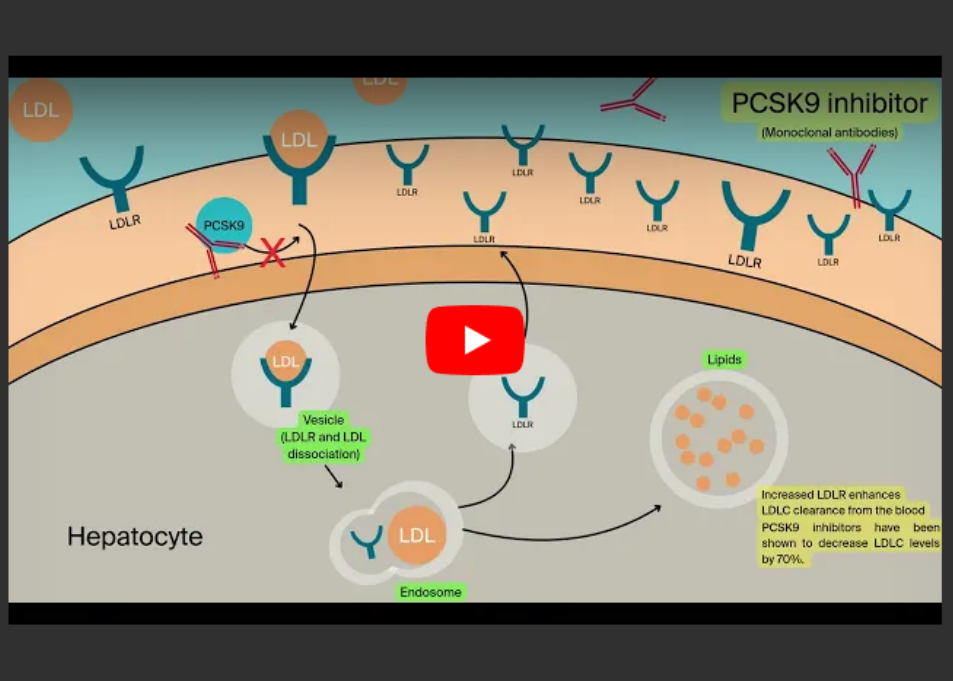
Examples of PCSK9 inhibitors: Alirocumab, Evolocumab (mab= monoclonal antibodies)

How do PCSK9-inhibitors work?

Proprotein convertase subtilisin kexin type 9 (PCSK9) is an enzyme predominantly produced in the liver. PCSK9 binds to the LDL receptor on the surface of hepatocytes. This binding facilitates the endocytosis of the LDL-R The LDL-R and PSCK9 complex is transported to the endosome Binding of PCSK9 also causes lysosomal degradation of LDL-R.



The video below describes the mechanism of action of PSCK9 inhibitors.


LDL
PCSK9 inhibitor
(Monoclonal antibodies)
LDLR
PCSK9
LDL
LDLR
Vesicle (LDLR and LDL dissociation)
Lipids
Hepatocyte
LDL
Endosome
Increased LDLR enhances LDLC clearance from the blood PCSK9 inhibitors have been shown to decrease LDLC levels by 70%.

PCSK9 inhibitors have been shown to decrease LDL-C levels by 70% and reduce TGs by 30%. They have also shown a significant decrease in cardiovascular event risk reduction by 50%. These agents are indicated for use in patients with familial hypercholesterolemia or have an established ASCVD risk

What are the potential ADRs?

PSCK9 inhibitors are available as SC injections. These agents can cause injection site reactions. **Hypersensitivity reaction** like rash, pruritis, and urticaria can occur with these injections.

Fibrates

Examples of Fibrates: Gemfibrozil, Fenofibrate

How do Fibrate work?

The peroxisome proliferator–activated receptors (PPARs) are members of the nuclear receptor family that regulate lipid metabolism. PPARs function as ligand-activated transcription factors. The endogenous ligands for PPARs are fatty acids or eicosanoids. Fibrates such as fenofibrate or gemfibrozil are PPAR agonists (or activators). Activation of PPARs leads to an increased expression of lipoprotein lipase, which leads to decreased TG concentrations.

PPAR activation by the fibrates also decreases apolipoprotein (apo) CIII expression. This leads to a decrease in VLDL formation and subsequently lowers TG levels. PPARs increase apo AI and apo AII expression, increasing HDL-C level.

Given that fibrates lower TG levels, they are often used to treat hypertriglyceridemia.

What are the potential ADRs?

GI effects mild GI disturbances are common with the use.

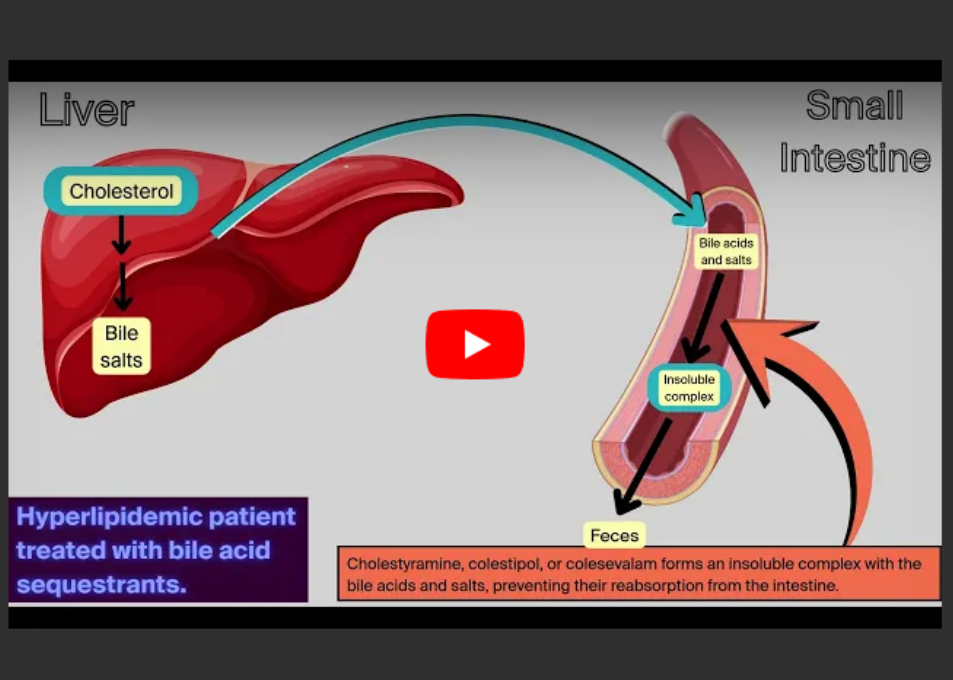
Skeletal/muscular: Myositis (muscle inflammation) can occur.

Gallstones: Fibrates can inhibit biliary secretions, leading to gallstone formation.

Bile acid sequestrants

How do bile acid sequestrants work?

Examples of bile sequ and colesevelam


Liver
Small Intestine
Cholesterol
Bile salts
Bile acids and salts
Insoluble complex
Feces
Hyperlipidemic patient treated with bile acid sequestrants.
Cholestyramine, colestipol, or colesevalam forms an insoluble complex with the bile acids and salts, preventing their reabsorption from the intestine.

What are the potential ADRs?

Common ADRs include GI disturbances such as constipation, nausea, and flatulence. Since these agents are resins, they can also interfere with absorption of fat-soluble vitamins (A,D, E, and K). They can also bind to some drugs such as Digoxin, Warfarin, and thyroid hormones and decrease their therapeutic efficacy.

Fish Oil

Examples: Icosapent ethyl, Docosahexaenoic and eicosopentaenoic acids

The mechanism of act not completely understood. The propo ion of VLDL and TG synthesis by increasing lipoprotein lipase activity. The omega-3 PUFAs eicosapentaenoic acid (EPA) and docosahexaenoic acid (DHA) are found in marine sources such as tuna, halibut, and salmon. While there are OTC supplements available, Icosapent ethyl is the only prescription product that contains only EPA and, unlike other fish oil supplements, does not significantly raise LDL-C.

What are the potential ADRs?

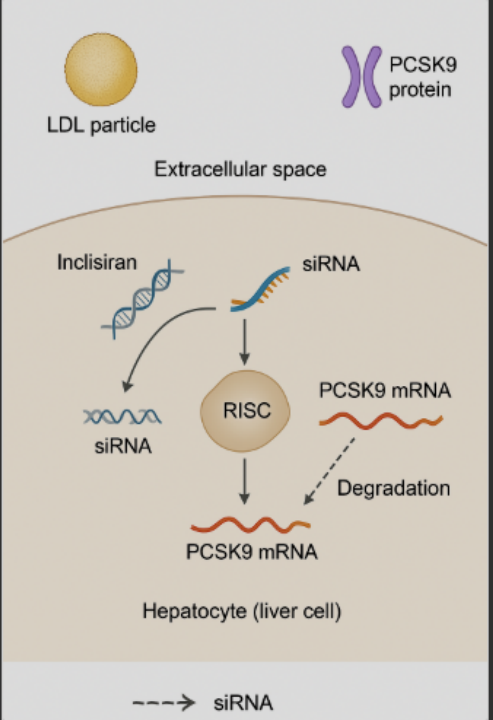
The most common adverse effects include GI effects of abdominal pain, diarrhea, and nausea. In addition, the fishy aftertaste is also a side effect from the use.

Evidence has shown that lowering LDL-C can reduce the risk of cardiovascular disease (CVD). The decision and choice of agent used to lower LDL-C incorporates the LDL-C and the patient's estimated 10-year CVD risk. The shared decision-making by discussing the management plan between the patient and provider is important for the success of the care plan.

Inclisiran

How does Inclisiran work?

Inclisiran is a double-stranded small interfering ribonucleic acid (siRNA) that prevents proprotein convertase subtilisin/kexin type 9 (PCSK9) translation in the liver. This increases LDL-C receptor recycling and expression on the hepatocyte cell, which increases LDL-C uptake and reduces LDL-C concentrations in the circulation.



What are the potential ADRs?

Injection site reactions, erythema, hypersensitivity reactions, arthralgia (5%), and bronchitis (4%) have been reported with use.

What are the non–pharmacological options for treatment and prevention? (LO#3)

For patients with hyperlipidemia, **lifestyle modification is the cornerstone of management**

- Weight loss for obese patients
- Strict glucose control for patients with diabetes
- Aerobic exercise
- Avoidance of concentrated sugars and alcohol
- Smoking cessation (smoking oxides LDL and makes it more atherogenic)
- Limiting carbohydrates
- Consume fish that has high levels of omega-3 fatty acids (salmon, tuna, etc)

CLINICAL CORRELATION

Omega-3 polyunsaturated fatty acids (PUFAs) can be considered as an adjunct to other lipid-lowering therapies for individuals with elevated triglycerides (≥500 mg/dL). Although effective for triglyceride lowering, in general, omega-3 PUFA supplementation has not been shown to reduce cardiovascular morbidity and mortality. An exception is icosapent ethyl, which has been shown to reduce the risk of cardiovascular events in secondary prevention patients or high-risk primary prevention patients when added to a statin.

CASE CONNECTION

BACK TO INTRODUCTION ↑

On examination, his vital signs were stable, with mild tenderness in proximal muscles but no swelling, erythema, or strength deficits. The clinical picture was most consistent with statin-associated myalgia, though hypothyroidism, vitamin D deficiency, and other medication effects remained in the differential. The plan was to order creatine kin sts, and vitamin D levels. Discontinuing mptoms in two weeks is also an option. Reinforcing the importance of lipid control and lifestyle modification. These include regular physical exercise, a heart-healthy diet, avoiding tobacco use, etc.

Summary

- Antihyperlipidemic drugs lower blood lipid levels, including LDL and cholesterol, to treat dyslipidemia and reduce ASCVD risk.
- Statins stabilize coronary plaques, reduce CRP levels, and reverse endothelial dysfunction, but can cause hepatic dysfunction, muscle symptoms, and increase diabetes risk.
- Ezetimibe inhibits cholesterol absorption in the intestine, reducing LDL levels, and can cause myalgias and diarrhea.
- Bempedoic acid inhibits ACL enzyme, decreasing cholesterol formation and LDL levels, with potential ADRs including hyperuricemia.
- Niacin can cause hepatotoxicity and flushing.
- Fibrates are PPAR-α agonists and can be used for treating hypertriglyceridemia.
- PCSK9 inhibitors, such as Alirocumab and Evolocumab, significantly decrease LDL-C levels and cardiovascular event risk, with injection site reactions as potential ADRs. They are indicated for treating familial hypercholesterolemia.
- Non-pharmacological management of hyperlipidemia includes lifestyle modifications like weight loss, exercise, and dietary changes.

Review Questio

1. Which of the following works to lower triglyceride levels by activating PPAR-α?

A. ○ Atorvastatin
B. ○ Evolocumab
C. ● Gemfibrozil
D. ○ Inclisiran

Hide Explanation

The correct answer is Gemfibrozil (C). Gemfibrozil acts by activating PPAR-α and activating lipoprotein lipase to decrease fatty acid synthesis and triglyceride synthesis. Choice (A) atorvastatin inhibits HMG-CoA and decreases LDL-C levels. Choice (B) Evolocumab is a PSCK9 inhibitor and does not act on PPAR-α. Choice (D) Inclisiran inhibits PSCK9 translation to help decrease LDL-C levels.

2. Which drug class stabilizes coronary plaques?

A. ● Statins
B. ○ Bempedoic acid
C. ○ PCSK9 in
D. ○ Fibrates
E. ○ Ezetimibe

Hide Explanation

The correct answer is statins (A). Statins help maintain fibrous cap integrity, preventing plaque rupture. Bempedoic acid (B) and PCSK9 inhibitors (C) primarily reduce LDL levels without direct plaque stabilization. Fibrates (D) lower triglycerides, not directly stabilizing plaques. Ezetimibe (E) reduces cholesterol absorption, not directly affecting plaque stability.

3. Which of the following can cause hepatotoxicity and is contraindicated in patients with hepatic disease?

A. ○ Alirocumab
B. ○ Ezetimibe
C. ○ Gemfibrozil
D. ● Niacin

Hide Explanation

The correct answer is patoxicity and is contraindicated in patients with severe liver disease. Choice (B) Ezetimibe is not contraindicated in hepatic disease. Choice (C) Gemfibrozil can cause hepatotoxicity but is not contraindicated in patients with severe hepatic disease.

4. Why is smoking cessation considered a lifestyle modification for patients with high ASCVD risk scores?

A. ○ Smoking increases effect of statins
B. ○ Smoking decreases circulating LDL-C
C. ○ Smoking increases uptake of cholesterol
D. ● Smoking oxides LDL and makes it more atherogenic

Hide Explanation

The correct answer is D Smoking oxides LDL and makes it more atherogenic. An increase in atherogenic LDL can increase the risk for atherogenic plaque, increasing ASCVD risk. Choice (A) is incorrect since statins' potency or efficacy is not altered by smoking. Choice (B) is incorrect as smoking does not change circulating LDL-C levels. Choice (C) is incorrect as smoking does not change the uptake of cholesterol. Cholesterol uptake is determined by the expression of LDL-R.

H k?

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How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Justify the use of antihyperlipidemic agents based on their mechanism of action and pharmacokinetic profile~~
02 ☑ ~~Recognize common adverse drug reactions for antihyperlipidemic agents~~
03 ☑ ~~Discuss the non-pharmacological options to treat dyslipidemia~~

Objective Confidence: 3/3